Sobol-Initialized Quality-Diversity Evolution of a Sub-50 g Titanium Karambit

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***Abstract*—We present G4 (v4.1.0), a clean-slate architectural upgrade of the Jarambit evolutionary knife-design engine that resolves the representation, physics, and optimization bottlenecks identified in the G3 v3.0.0 retrospective audit while keeping every RS-karambit-v3 load magnitude and material assumption intact. Whereas G3 evolved over 179 genes (31 of them inactive) and produced a 57.02 g scored (59.34 g print mesh) part with two structural safety-factor shortfalls (solid tip** SF = 0.55 **under a 1.82 N·m twist about the origin-to-point axis and ring-clamped cantilever grip spine** SF = 0.515 **under a 3,427 N spine load), G4 introduces a declarative 130-gene schema (genome/G4.0.toml, zero dead genes, zero hand-authored anchor) coupled to a true-3D curvilinear implicit kernel, a compliant four-finger Winkler hand-contact equilibrium model (HM-1), and a continuous quality-diversity objective (W_ERGOLIGHT_v2, hard_gates = false) over a Lloyd-relaxed CVT archive with a four-emitter black-box bandit (Sep-CMA-MAE, DE/current-to-pbest/1/bin, SBX+PM, and Iso+LineDD) initialized 100 % from scrambled Joe–Kuo Sobol samples. Every release check in G4ReleaseAdmissionContract is fail-closed: three-level Roache grid-convergence index (**$\mathrm{GCI}_{21} = 0.27\,\% \le 5\,\%$**), per-load-case failure sequence ledger (**$\min \mathrm{seq} = 1.68 \ge 1.10$ **across lateral, torsion, and 3,427 N spine first-yield), frame-invariant TET10 solid-tip FE under the full 1.82 N·m tip twist with a local tip-bisector diamond crown (**$\mathrm{SF}_{\mathrm{LCB}} = 8.26 \ge 1.25$**), 2D quadratic T6 serration tooth checks across all teeth including end teeth (**$6/6$ **inner and** $5/5$ **spine teeth,** $\mathrm{SF}_{\mathrm{LCB}} = 7.50 \ge 1.25$**), single-$k_t$ Goodman fatigue (**$\mathrm{SF}_{\mathrm{LCB}} = 8.39 \ge 1.10$ **at 300 MPa endurance limit), and resolution-invariant morphological erosion powder drainability. Across a multi-seed Sobol-initialized cohort and five-arm ablation study (E0–E5), full G4 (E3) achieves a median score of** 94.15 **(IQR** 0.18**) and selects winner `g4v410_s042_a19374a8` at 37.87 g finished mass (38.15 g with bevel grind stock; 63.8 % hollowness vs. its 104.58 g solid envelope), scoring 94.49 under W_ERGOLIGHT_v2 (98.15 on legacy W_ERGOLIGHT_v1) with zero constraint violations (**$V(x) = 0$**) across all 29 targets.**

***Index Terms*—Constrained quality diversity, FI-2Pop, MAP-Elites, CMA-MAE, additive manufacturing, Ti-6Al-4V, finite elements, Roache GCI, Winkler foundation.**

1 Introduction and Retrospective Audit of G3

Designing a one-piece, hollow Ti-6Al-4V karambit for laser powder-bed fusion (L-PBF) requires simultaneous satisfaction of coupled geometric, ergonomic, structural, and additive-manufacturing constraints. In our preceding G3 v3.0.0 release, an eight-island MAP-Elites search evolved a double-edged, double-serrated claw blade with an internal triply periodic minimal surface (TPMS) lattice. A post-release red-team audit of the G3 codebase, physics pipeline, and release receipt exposed six fundamental engineering lessons that motivate G4:

1) **Soft scalarization permits structural shortfalls.** In W_ERGOLIGHT_v1, all 21 objective terms were combined linearly through a soft knee function sfk without a feasibility barrier. Because mass carried weight $w = 0.370$ while individual structural terms carried 0.0174–0.0700, the optimizer rationally sacrificed safety factors on the solid tip (SF = 0.431) and spine-loaded grip (SF = 0.515) whenever thinning the geometry gained more mass points than it lost in structural points.
2) **Boundary-condition mismatch on the grip and tip.** G3 evaluated the 3,427 N spine baton impulse on a hollow grip cantilevered from the pommel ring ($x \approx 9.9$ mm), producing an unphysical ∼55 mm moment arm that does not exist when the knife is held in a closed fist supported by the palm and four fingers [12], [13]. Similarly, G3 applied thefull 1.82 N·m body torque to the distal 3 mm of the point about the global origin-to-tip chord rather than the local tip bisector.
3) **Coupled point planform and spine thickness.** In G3, the distal width profile $w(s)$ blended into the body width via a polynomial ramp over 4 mm, coupling the 3 mm planform point angle ($\approx 92^\circ$ on the v3 winner) to the spine thickness knots and preventing an acute $\le 55^\circ$ point from coexisting with a torsion-resistant spine.
4) **Surrogate fidelity gaps.** In G3's search loop, the plate mesh mesh_domain(&b_outer, &[], 0.9) ignored through-blade fuller windows, the three-level Roache grid-convergence index (GCI) [6] was held at a constant placeholder 0.50, the Goodman fatigue formula divided by capacity instead of multiplying by unit stress, the serration check evaluated only $N - 1$ interior teeth between N scallops, and voxel flood-fill drainability at $h = 0.40$ mm passed single-voxel diagonal necks that close at finer print resolutions.
5) **Genome bloat and representation ceiling.** Of the 179 genes in G3.0, 26 genes (including legacy CPPN weights and fixed parameters) were never read by the decoder, 5 self-adaptive σ genes were hashed into genome_sha256, and the blade mid-surface was strictly planar ($z \equiv 0$).
6) **Uniform-skin mass floor.** Using a uniform ∼1.0 mm grip skin, an unperforated longitudinal web, and a solid pommel ring rim imposed a ∼60–68 g mass floor.

G4 (crates/g4) is engineered from the ground up to resolve every one of these findings while preserving the historical v1/v2 and v3.0.0 artifacts byte-for-byte intact.

2 Declarative G4.0 Genome and True-3D Implicit Kernel

Release v4.1.0, schema hash `cfbe4e93`, decoder hash `8300db0c`. Every measured number in this paper is generated from evidence/g4/v4.1.0/release.json and evolution_summary.json by g4-release-tex.

2.1 Single-Source Declarative Schema (genome/G4.0.toml)

G4 replaces hand-maintained index arrays with a single declarative schema genome/G4.0.toml compiled into Rust via build.rs. The schema defines **130 live phenotype genes** (116 continuous F64 and 14 discrete/categorical I64 genes; **zero dead genes** and **zero search-strategy genes** in the phenotype hash) partitioned into nine anatomical modules (Table 1):

- **Module H (Hand & Grip, 24 genes):** Anthropometric hand scale, superelliptic grip cross-sections, palm swell, finger choil depths, and flange-graded skin thickness law $t_{\text{skin}}(\theta, u) = t_{\text{base}} + c_2 \cos(2\theta) + s_1 \sin\theta + a_x(u - 0.5)$ (**Lever L-1**).
- **Module G (Ring & Guard, 16 genes):** Ergonomic finger-ring lip crown, neck fillet radius, drained hollow ring gallery fraction ring_hollow_t (**Lever L-4**), and parametric B-spline guard contours.
- **Module P (Blade Planform, 16 genes):** Log-spiral curvature progression, chord reach target, and a decoupled linear-wedge point planform law over $[0, L_{\text{tip,plan}}]$ parameterized directly by the true planform point angle $\theta_{\text{tip}} \in [32°, 48°]$.
- **Module Z (3D Curvilinear Camber & Twist, 8 genes):** Out-of-plane blade mid-surface camber $z_c(s)$ and axial pre-twist $\psi(s)$ framed via parallel-transport (Bishop) frames along the 3D spine curve $\gamma(s) \in \mathbb{R}^3$.
- **Module X (Blade Section, Bevels & Crown, 18 genes):** Core thickness, bi-concave fuller grooves (**Lever L-5**), primary bevel angles (17.5°–22.5° per side), apex flats (0.05–0.11 mm), and the distal tip-bisector diamond crown reinforcement t_tip_crown_mm over $a \in [1.5, L_{\text{crown}}]$ mm (**Lever L-8**).
- **Module I (Internal Lattice, Webs & Bulkheads, 18 genes):** Spatially graded Gyroid/Schwarz-D Sheet-TPMS relative densities (**Lever L-3**, **L-6**), perforated Warren web hole fraction rib_hole_frac (**Lever L-2**), transverse finger-station bulkheads, and powder-removal port diameters (≥ 2.50 mm).
- **Module E (Dual-Edge Serrations, 12 genes):** Scallop counts, depths, gullet duty factors, and plunge offsets on both inner and spine cutting edges.
- **Module T (Alien Trabecular & Exoskeletal Architecture, 10 genes):** Continuous gate alien_gate, tri-arched Bishop-frame neck-to-blade trabecular struts, and neutral-axis flank windowing.
- **Module A (Zero-Mean Surface Texture, 8 genes):** Knurl/Voronoi/flute blend weights and amplitude (0.10–0.38 mm) enforced as a strictly zero-mean displacement ($\int h_{\text{tex}}\, dA = 0$, **Lever L-7**).

2.2 Decoupled Point Wedge and Tip-Bisector Diamond Crown

In G4, the planform half-width $w(a)/2$ over the distal tip zone $a \in [0, L_{\text{tip,plan}}]$ follows an exact linear wedge $w_{\text{wedge}}(a) = 2r_{\text{tip}} + 2a\tan(\theta_{\text{tip}}/2)$ before transitioning via a C^2 quintic smoothstep ($\tau^3(10 - 15\tau + 6\tau^2)$ over $a \geq 3.2$ mm) into the body width law. Consequently, the planform point angle at both $a = 1.0$ mm and $a = 3.0$ mm equals θ_{tip} (37.90° $\leq 55°$ on the winner) independently of spine thickness. Meanwhile, the spine thickness over $a \in [1.5, L_{\text{crown}}]$ receives a local diamond-ridge crown $\Delta t_{\text{crown}}(a) \propto \sqrt{\sin(\pi u_c)}\, t_{\text{tip,crown}}$, raising the polar moment of inertia $J \propto w\, t_{\max}^3$ by $> 2.4\times$ at the cost of only +0.19 g of distal titanium (evidence/g4/studies/e06_tip_crown.json).

Table 1 | Declarative G4.0 genome modules (130 live phenotype genes, zero dead genes)

Mod.	Anatomical Subsystem	Cont.	Disc.	Primary Mass / Structural Role
H	Hand & Flange-Graded Grip	24	0	Lever L-1, L-7 (I_{zz} +38 %, flank relief)
G	Drained Ring & Guard	16	0	Lever L-4 (hollow ring gallery, fillet K_t)
P	Log-Spiral Planform & Point	16	0	Gate S4 ($\theta_{\text{tip}} \leq 55°$ at 1 & 3 mm)
Z	3D Camber & Twist Frame	8	0	Bishop parallel-transport 3D mid-surface
X	Section, Fuller & Tip Crown	18	0	Lever L-5, L-8 (diamond tip crown)
I	TPMS, Warren Web & Ports	14	4	Lever L-2, L-3, L-6 (perforated web, drain
E	Dual-Edge Serration Law	10	2	All teeth checked (N/N including ends)
T	Trabecular Exo-Architecture	6	4	Tri-arch Guard-Neck bridge, flank windov
A	Zero-Mean Grip Texture	4	4	Lever L-7 ($\int h_{\text{tex}}\, dA = 0.0$ g)
Total (G4.0.toml)		**116**	**14**	**130 live genes (schema_sha256: `cfbe`**

3 Requirements, Hand Contact Model, and Continuous Objective

3.1 Machine-Readable Specification (RS-karambit-v4.toml)

All 29 structural, geometric, and manufacturing targets, material constants, and ergonomic target bands are loaded from RS-karambit-v4.toml (hard_gates = false) with `#[serde(deny_unknown_fields)]` and traceable provenance in PROVENANCE.md, keeping every load magnitude and material assumption from RS-karambit-v3: 1.82 N·m solid-tip twist, 3,427 N spine load judged on first yield (σ_y = 828 MPa), and stress-relieved L-PBF Ti-6Al-4V fatigue endurance limit (σ_e = 300 MPa at 10^7 cycles before thin-wall $k_t(t)$ applied once). For each structural target $j \in \{1, \ldots, 17\}$, G4 evaluates the uncertainty-bounded lower confidence bound (LCB) safety factor:

$$\text{SF}_{j,\text{LCB}}(x) = \frac{\text{SF}_j(x)}{1 + k_{\text{LCB}}\sqrt{\text{GCI}_j^2 + \delta_{\text{mat}}^2}}, \qquad k_{\text{LCB}} = 1.645, \quad (1)$$

where GCI_j is the measured three-level Roache grid-convergence index and $\delta_{\text{mat}} = 0.022$ is the L-PBF Ti-6Al-4V yield scatter [14].

3.2 Unified Winkler Hand Contact Equilibrium (HM-1)

Rather than clamping the pommel ring in space while applying a 3,427 N spine load to the blade, G4 models the closed-fist human grip as a four-station compliant Winkler foundation (eval_g4_hand_model_hm1) at the little/ring, middle, index-groove, and index-choil/palm-heel stations $x_i \in \{18, 39, 60, 81\}$ mm [12], [13]. Solving the 2×2 rigid-body Winkler equilibrium for each external wrench ($\mathbf{F}_{\text{ext}}, \mathbf{M}_{\text{ext}}$) satisfies force and moment balance to $< 10^{-12}$ N and distributes the load across the inter-finger saddle span ($l_{\text{HM1}} \approx 14.5$ mm), eliminating the artificial 55 mm ring-clamp cantilever lever arm and raising the first-yield spine sequence ratio from 0.515 (v3 ring clamp) to ≥ 1.62 (evidence/g4/studies/e01_v3_rescore.json).

3.3 Continuous Multi-Objective Scalar (W_ERGOLIGHT_v2)

Per the owner's v4 specification (hard_gates = false), G4 evaluates a smooth, continuous objective over $[0, 100]$ with no step-function gates:

$$W_{\text{ERGOLIGHT_v2}}(x) = 100\,E(x)\,(0.370\,L_{\text{mass}}(x)+0.630\,\bar{S}_{\text{struct}}(x))\,\psi_{\text{smooth}}(x) \tag{2}$$

where $L_{\text{mass}}(x) = \text{clip}((90 - m_{\text{finished}}(x))/(90 - 28), 0, 1)$ rewards low finished mass, $\bar{S}_{\text{struct}}(x) = \frac{1}{29}\sum_{j=1}^{29} \text{sfk}(r_j(x))$ evaluates the RS-karambit-v3 saturation curve $\text{sfk}(r) = 0.94\min(r, 1) + 0.06(1 - e^{-\max(r-1,0)/0.15})$ across all 29 targets, $E(x) = \prod_{k=1}^{8} e_k(x)^{\beta_k}$ is the eight-factor anthropometric ergonomic geometric mean, and $\psi_{\text{smooth}}(x) = \exp(-3.5\sum_j v_j(x) - 12.0\sum_j v_j(x)^2)$ is a smooth boundary-steepened factor that equals 1.0 whenever all targets are met ($v_j(x) = \max(0, 1-r_j(x)) = 0$) and decays continuously without step discontinuities.

4 Calibrated Multi-Fidelity Physics and Verification

Every surrogate and Tier-2 solver in crates/g4/src/physics.rs is verified by automated tests in vv_suite.rs and R&D study receipts in evidence/g4/studies/:

1) **Sub-voxel smoothed-Heaviside volume integration (T-MASS-1, Study E-02):** Evaluating $H_\epsilon(\phi)$ with 8-point Gauss-Legendre sub-cell quadrature where $|\phi| \le \sqrt{3}h/2$ achieves 0.002 % volume error on a $R = 10$ mm sphere and 0.044 % on a torus at $h = 0.40$ mm (< 0.25 % drift between $h = 0.40$ mm and $h = 0.20$ mm).
2) **Resolution-invariant erosion drainability (T-DRAIN-1, Study E-02):** Cavity air is eroded by the physical powder-clearance ball radius $r_{\text{erode}} = 0.50$ mm ($\phi_{\text{void}} \le -(r_{\text{erode}} + 0.5h)$) before 6-connected flood fill, guaranteeing identical drainability classification across $h \in \{0.40, 0.20, 0.10\}$ mm.
3) **Hole-aware plate FE and 3-level Roache GCI (T-PLATE-1..2):** Through-blade fenestrations are subtracted from the CST+DKT sandwich plate mesh [7], unreferenced interior hole vertices are locked, and lateral capacity is evaluated across three self-similar finite-element mesh refinement levels (16 × 4, 24 × 6, and 36×9, refinement ratio $r = 1.5$) to compute a real Roache $\text{GCI}_{21} = 0.27$ % (observed order $p = 1.72$) admitted by PlateGciAdmission [6].
4) **Frame-invariant TET10 solid-tip FE (T-TIP-1..3, Study E-06):** The distal 28 mm tip submodel is transformed into the local tip-bisector frame with cell-centred thickness sampling and evaluated under LC-TIP-1 (the full 1.82 N·m RS-karambit-v3 tip twist) and LC-TIP-2 (60 N lateral tip pry), achieving 0.000 % rigid-rotation variance.
5) **All-tooth T6 serration evaluation and single-k_t Goodman fatigue (T-SERR-4):** Every serration tooth $i \in \{0, \dots, N-1\}$ on both edges—including both end teeth—is evaluated under combined 180 N snag and 60 N lateral pry with Peterson notch sensitivity K_f [11] and local thickness knockdown $k_t(t_{\text{local}})$, while Goodman fatigue at $R = 0$ is computed from true root bending stress $\sigma_{\max} = 180\,\sigma_{1N}$ against the 300 MPa endurance limit with $k_t(t)$ applied once.
6) **Locking-free B-bar hex8 voxel FE (T-VFE-1..4):** Selective-reduced / Wilson-condensed B-bar brick elements [8] pass the constant-strain patch test to 1.1×10^{-16} and reduce thin-beam bending locking error from 85.9 % (plain hex8) to $< 10^{-12}$.

5 Sobol-Initialized Quality-Diversity Search

crates/g4/src/search.rs implements a deterministic, batch-synchronous quality-diversity engine with zero hand-authored anchor genomes and zero named-gene repair rules:

- **100 % Scrambled Joe–Kuo Sobol Initialization:** Every initial individual across all five experimental arms (E0–E5) is generated purely from scrambled Joe–Kuo Sobol quasi-random sequences (G4Genome::from_sobol) [15]. No default_anchor or default_baseline constructor exists in crates/g4.
- **Lloyd-relaxed 6D CVT archive:** Niche centroids in the 6D phenotypic descriptor space (finished mass m, blade reach R, claw turn angle $\Delta\theta$, point angle θ_{tip}, grip H/T, and hollowness fraction η_{void}) are computed via deterministic Lloyd relaxation [2].
- **UCB1 four-emitter black-box bandit:** Offspring batches are generated by four 100 % gene-agnostic black-box continuous variation operators scheduled by UCB1: (1) Sep-CMA-MAE [3] initialized from the top Sobol archive elites, (2) DE/current-to-pbest/1/bin differential evolution over archive elites, (3) simulated binary crossover (SBX) with polynomial mutation, and (4) sparse-coordinate Iso+LineDD directional crossover.
- **Periodic $h = 0.20$ mm Tier-2 verification and cross-arch determinism:** Top archive elites are periodically re-evaluated at Tier 2 ($h = 0.20$ mm, 3-level plate GCI, and 3D TET10 tip FE). All transcendental math uses pure-Rust libm and parallel batches are reduced in deterministic index order, guaranteeing bit-identical results across thread counts (T-DET-1).

6 Experimental Results, Mass Waterfall, and Verified Winner

6.1 Eight-Lever Mass Waterfall (Study E-05)

Table 2 traces the sequential application of the eight structural mass-reduction levers (L-1–L-8) starting from a uniform-skin, unperforated baseline (L-0, 56.53 g finished). Together, flange-graded grip skins (L-1), perforated Warren webs (L-2), spatially stress-graded TPMS (L-3), the drained hollow ring gallery (L-4), the bi-concave blade fuller (L-5), decoupled hollow blade core (L-6), lean anthropometric grip envelope with zero-mean texture and trabecular flank windows (L-7), and the distal tip-bisector crown (L-8) reduce finished mass by **12.24 g** (−21.7 %, down to **44.29 g**) while maintaining positive safety margins across all 29 targets.

6.2 Multi-Seed Cohort and Five-Arm Ablation Study (E0–E5)

Table 3 summarizes the five-arm experimental cohort executed by g4-evolve (evidence/g4/v4.1.0/evolution_summary.json). Full G4 (E3, 5 seeds) achieves a median score

Table 2 | Eight-lever mass reduction waterfall (evidence/g4/studies/e05_mass_levers.json)

Step	Architectural Lever Applied	Mass (g)
L-0	Uniform-skin unperforated baseline	56.53
L-1	Flange-graded grip skin (0.80 mm flanks, +0.42 mm chords)	53.40
L-2	Perforated Warren web (62 % circular cutouts)	51.64
L-3	Spatially stress-graded Sheet-TPMS core	47.99
L-4	Drained hollow ring gallery + 32.0 mm ring OD	47.86
L-5	Bi-concave blade fuller + lean 73.5 mm chord reach	45.57
L-6	Decoupled hollow blade core ($\rho_{\text{blade}} = 0.13$)	44.27
L-7	Lean anthropometric grip + zero-mean texture + Mod. T	44.41
L-8	Distal tip-bisector diamond crown (+0.52 mm)	**44.29**

Table 3 | Multi-seed cohort and ablation comparison (evidence/g4/v4.1.0/evolution_summary.json)

Experiment Arm	Seeds	Med. Score	IQR	Med. Mass (g)
E3: G4 Full (Continuous QD + Mod. T)	5	**94.15**	0.18	**39.27**
E5: G4 Core (No Module T)	3	94.09	—	39.16
E2: Single-Pop Sep-CMA-ES	3	94.08	—	—
E1: Legacy W_ERGOLIGHT_v1	3	92.95	—	—
E0: Pure Scrambled Sobol	3	91.36	—	—

of **94.15** (IQR 0.18) and median finished mass of **39.27 g**, outperforming the no-Module-T ablation (E5, median 94.09, mass 39.16 g), single-population CMA (E2, median 94.08), legacy W_ERGOLIGHT_v1 search (E1, median 92.95), and pure scrambled Sobol search (E0, median 91.36).

6.3 The Sub-50 g Winner (`g4v410_s042_a19374a8`)

Table 4 compares the Tier-2 verified G4 v4.1.0 winner `g4v410_s042_a19374a8` (genome hash `a19374a8`) against the G3 v3.0.0 winner hz4i013_g00995_d206e3c9. The G4 winner weighs **37.87 g** finished (**38.15 g** with the +0.30 mm bevel grind stock, vs. G3's 57.02 g scored / 59.34 g print mesh), partitioned across ring 4.40 g, grip 20.03 g, guard 2.09 g, and blade 11.34 g (root 2.65 g, mid 4.31 g, tip 4.37 g). Every one of the 17 structural LCB safety factors meets or exceeds its required target ($V(x) = 0$).

Table 4 | Head-to-head comparison: G3 v3.0.0 winner vs. G4 v4.1.0 winner `g4v410_s042_a19374a8`

Metric / Target	Req. (v4)	G3 v3.0.0	
Finished mass (g)	< 50.0	57.02 (scored)	
With grind stock (g)	—	59.34 (print mesh)	
Solid envelope mass (g) / Hollowness	—	101.0 / 41.2 %	**104.5**
Ring / Grip / Guard / Blade mass (g)	—	9.0 / 28.6 / 2.4 / 19.7	**4.40 / 20.03 / 2**
Score (W_ERGOLIGHT_v2 / legacy v1)	—	— / 80.78	**94**
Platform pull angle at 3 mm (°)	≤ 55.0°	92.22°	
Radial 3-level plate GCI_{21} (%)	≤ 5.0 %	non-monotone	
Hilt-lateral 300 N SF_{LCB}	≥ 1.35	1.43	
Hilt-lateral 260 N SF_{LCB}	≥ 1.15	0.99	
Body torsion 1.82 N·m SF_{LCB}	≥ 1.25	12.69	
Human-max torsion 16.2 N·m SF_{LCB}	≥ 1.10	1.43	
Solid-tip twist 1.82 N·m SF_{LCB}	≥ 1.25	0.55	
Solid-tip lateral pry 60 N SF_{LCB}	≥ 1.20	—	
Spine push 500 N SF_{LCB}	≥ 1.50	6.22	
Spine load 3,427 N first-yield SF_{LCB}	≥ 1.10	0.52 (ring clamp)	
Ring bearing 300 N / back 30 N SF_{LCB}	≥ 2.00	5.66 / 3.92	**18**
Grip bilateral squeeze 250 N SF_{LCB}	≥ 1.50	—	
Serration tooth snag+pry SF_{LCB} (checked)	≥ 1.25	2.13 (4/5, 5/6)	**7.50**
Goodman high-cycle fatigue 180 N SF_{LCB}	≥ 1.10	1.06 (single k_t)	
Lateral-torsional buckling λ_{LCB}	≥ 2.00	3.85	
Combined 70 % lat. + 70 % twist SF_{LCB}	≥ 1.15	—	
Per-case failure sequence min ratio_{LCB}	≥ 1.10	0.52 (spine)	
TPMS face-sheet wrinkling SF_{LCB}	≥ 1.35	—	
CoM distance to index choil (mm)	≤ 15.0	11.40	

7 Scope, Limitations, and Reproducibility

Like G3, G4 is a computational engineering release: physical L-PBF fabrication, stress relief, CNC bevel grinding, and destructive mechanical testing remain required before operational qualification. All code, study receipts (evidence/g4/studies/), cohort telemetry (evidence/g4/v4.1.0/), and the six-file manufacturing upload kit (cad/) are deterministically reproducible via cargo test --manifest-path crates/g4/Cargo.toml --release and g4-studies, g4-evolve, g4-release-tex, and g4-export.

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